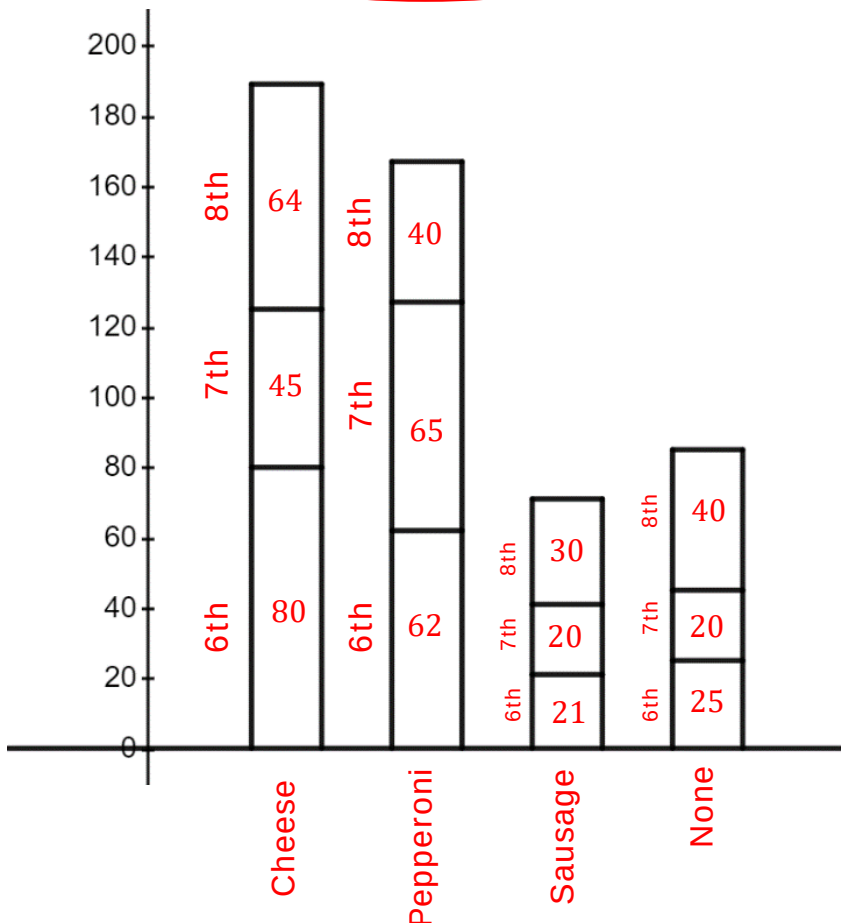


Southside Middle School conducted a school pizza party to celebrate the 100th day of school. Each student was given a choice of cheese, pepperoni, or sausage pizza. Some students did not want pizza. The results are shown in the frequency table below. A segmented bar graph was created with the data from the table. Use the figures to complete questions 1-5.

	6 th	7 th	8 th	Total
Cheese	80	45	64	189
Pepperoni	62	65	40	167
Sausage	21	20	30	71
None	25	20	40	85
Total	188	150	174	512

1. Circle the marginal frequencies in the table.

2. Box the joint frequencies in the table.



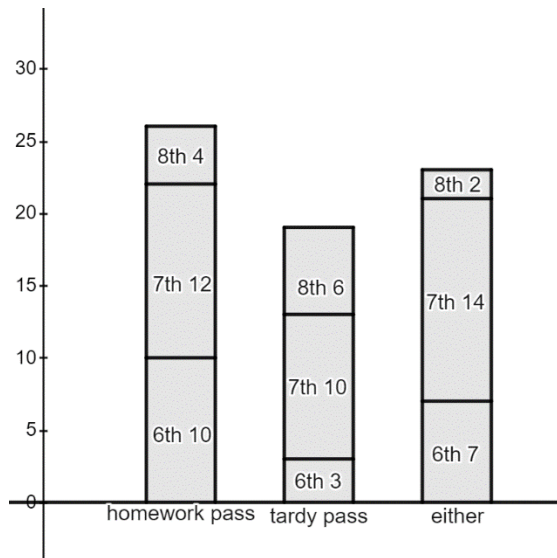
3. Label each marginal frequency in the boxes of the segmented bar graph.

4. Label the value for each joint frequency on the segmented bar graph.

5. Based on the data what type of pizza is the least favorite for each grade level?

Sausage

A teacher at a local middle school is deciding on an incentive for good behavior. The teacher polled students from each grade level to determine if students would prefer a tardy pass, homework pass, or either. The results are included on the segmented bar graph below. Use the bar graph to complete questions 6-10.



6. What does the joint frequency value of 12 represent on the bar graph?

The amount of 7th graders who prefer receiving a homework pass

7. How many 7th graders would prefer a tardy pass?

10

8. How many total students would prefer a homework pass?

26

9. How many 7th and 8th graders would like either reward?

16

10. If the teacher makes her decision based on the results in the segmented bar graph, what should she do? Explain your reasoning.

Homework pass because it is the most popular choice for 6th and 7th grade if you consider the votes for "either," it makes it a comparable choice for 8th grade as well.